

LEDGER · 2026-08-27 · BIOMANUFACTURING / SYNTHETIC BIOLOGY

Commercial Bacteriorhodopsin Production via Non-Sterile Hypersaline Cultivation of *Halobacterium salinarum*

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Inorganic Optically Variable Pigments (Merck Iriodin or multichrome chameleon pigments)
Entry application	High-Security Anti-Counterfeiting Printing Inks
Measured advantage	none
Time to first revenue	19 months
Regulatory risk	Low — no material regulatory CapEx reported (\$0 estimated)

Why it failed

- Voided by: Capital and time to revenue, Economics undisclosed, Form-factor / handling regression

Assessment

The proposal fails gross margins, clocking in at 58.5% at price parity which falls below the mandatory 70% threshold. Form factor handling presents a clear regression over inert OVPs due to hygroscopic biological powder requirements and solvent incompatibility. Time to first paid delivery is explicitly estimated beyond 18 months due to binder formulation hurdles.

Also flagged

- Duplicate references inflating the evidence base
- runsheet_no_quantities

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-27 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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